

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	29 JUNE 2025
Team ID	
Project Name	Human Development Index (HDI) Predictor
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication	User Login, User Registration (Optional), Session Management
FR-2	HDI Data Input	Enter Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, GNI per Capita, Country Selection
FR-3	HDI Prediction	Predict HDI Score using Trained Machine Learning Model
FR-4	Visualization & Reporting	Display HDI Score, HDI Category, Graphs, Country Comparison, Download Report

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	User-friendly and responsive web interface for easy HDI prediction.
NFR-2	Security	Input validation, secure request handling, and protection against invalid user inputs.
NFR-3	Reliability	Generate consistent and accurate HDI predictions using a trained machine learning model.
NFR-4	Performance	Provide prediction results and visualizations within a few seconds.
NFR-5	Availability	The Flask web application should be available whenever the local server is running.
NFR-6	Scalability	The system should support additional datasets, countries, and future machine learning models without major modifications.